

EDUCATION

HEC Lausanne

M.Sc. Finance (Quantitative), GPA: 4.9/6.0

Lausanne, Switzerland

2019 – 2021

- **Thesis:** Rough Volatility Modeling in option pricing (6.0/6.0)
- **Coursework:** Econometrics, Mathematics, Python/C++, Statistics

HEC Lausanne

B.Sc. Economics, GPA: 4.6/6.0

Lausanne, Switzerland

2016 – 2019

WORK EXPERIENCE

MyLift

Founding Engineer

Dublin, Ireland

February 2026 – Present

- Led end-to-end development of an Elixir/Phoenix medical tourism platform connecting French-speaking patients (France, Belgium, Switzerland) with 8 verified Tunisian doctors and 5 clinics across 10+ procedures and specialties. Owned architecture and product strategy.
- Integrated Stripe for split payments across multiple care providers and Daily.co for real-time video consultations, enabling cross-border transactions between European patients and care providers in Tunisia.
- Built role-based dashboards for doctors, clinics and patients along with an internal admin tool that powers the founding team's onboarding workflow for healthcare providers (clinics and doctors).
- Implemented direct client-to-S3 uploads via pre-signed URLs for galleries, profile images and leveraged a CDN to optimize media delivery performance.

Millennium

Software Engineer

Dublin, Ireland

December 2025 – February 2026

- Initiated automation of CI runner deployment using SaltStack, designed for easy rollout by adding a single configuration file to GitLab allowing a team of 30 people to initialize individual runners.
- Added change-data capture for configuration edits, giving the team an audit trail of who modified what and when on Virtual machine, network and cluster configurations.

Gartner

Software Engineer

Barcelona, Spain

January 2024 – November 2025

- Developed a GraphQL Elixir API serving over 100 million users annually, implementing a Query Specification pattern across PostgreSQL and Elasticsearch for composable query construction and consistent pagination.
- Rebuilt ad monetization interface, consolidating fragmented data sources and eliminating inconsistencies that disrupted ad delivery. This interface supports 100\$M+ in annual revenue.
- Cut deployment times by 25% using ECR layer caching instead of brute caching.
- Built query complexity framework letting backend teams validate algorithm changes pre-rollout, preventing client breakage and doubling iteration speed on a previously rarely-touched algorithm.

Amazon

Software Developer

Iasi, Romania

May 2022 – October 2023

- Built a Python application to query product attributes (ASINs, etc.) on AWS Redshift, streamlining manual review for a 20 person team.
- Created a Python application enabling a team of associates to verify item inventory within fulfillment centers (10 people).
- Developed a Python application to automate the generation of batch vendor invoices from Excel spreadsheets, reducing redundant tasks for a team of associates working on billings (5 people).

PROJECTS

LiveDisplaced: Data visualization platform built upon Python and AWS services (ECS, Batch) to track global refugee movements from the United Nations data sources.https://github.com/EM51641/livedisplaced_

Async-sendgrid: Developed and maintained async-sendgrid, an open-source Python client for non-blocking email delivery on SendGrid, with 400,000+ PyPI downloads.<https://pypi.org/project/sendgrid-async/>

LANGUAGES

French, Arabic (native); English (C1); Spanish (B1)

SKILLS

Languages: Elixir, Python, TypeScript/JavaScript, Java, Nix

Backend & Data: Phoenix/LiveView, GraphQL, Postgres, Elasticsearch, Kafka, Redshift

Infra & Cloud: AWS (ECS/Fargate, EKS, Lambda, Aurora, S3, Batch), Terraform, Docker, Kubernetes

CI/CD & Observability: GitLab CI, GitHub Actions, SaltStack, Datadog, Prometheus, Grafana, OpenTelemetry